

Jacob Tang

408-482-7035 | jake.tang@gmail.com | [LinkedIn](#) | [Github](#)

EXPERIENCE

Software Engineer Intern

May 2025 – July 2026

Resmed

San Diego, CA

- Shipped a new **REST API endpoint** for generic device data ingestion and validated via **E2E testing** across 9+ environments.
- Strengthened system reliability for remote patient-device settings updates by designing **Java Cucumber test suites** covering untested management API endpoints.
- Improved **SNS/SQS observability** by designing and implementing self-service **Datadog monitoring**, backed by a versioned **Terraform** module supporting 5 monitor types.
- Reduced test data maintenance overhead by **20%** by replacing **10+ static JSON fixtures** with reusable, parameterized templates backed by **MongoDB** and **WireMock**.
- Reduced query latency and enabled low-downtime schema evolution for device data by designing **MongoDB document models** and implementing **Flyway-managed migrations**.
- Built a starter library to propagate and extract trace context for **JMS messages** across **ActiveMQ brokers**, enabling distributed tracing.
- Helped develop a **Github** workflow enabling teams to independently provision their own **SNS/SQS** resources, eliminating manual **AWS Console** setup.

Founding Backend Engineer

Feb 2025 – May 2025

BarBuddy

San Diego, CA

- Built the backend of a mobile app within a 3-person team, designing scalable **REST APIs** with **Django REST Framework** and **PostgreSQL** to serve app data to the frontend.
- Launched the application to **10+ users**, enabling users to discover and coordinate bar outings through a centralized mobile experience.

Information Technology Intern

May 2024 – Aug 2024

Natera

San Carlos, CA

- Imaged and configured **200+ laptops** for new hires, ensuring the installation of essential software.
- Performed SQL queries to retrieve and analyze data from over **1,000 Jira tickets** stored in Snowflake.

EDUCATION

San Diego State University

May 2026

Bachelor of Science in Computer Science

GPA: 3.9

- **Relevant Coursework:** Operating Systems, Cloud Computing, Machine Learning, Unix System Administration

PROJECTS

AI Tutor | *Python, FastAPI, Docker, Typescript, React*

- Developed a full-stack AI-powered educational platform that summarizes user notes directly from Google Drive using **MCP Server Integration**.
- Engineered RESTful APIs supporting user authentication, course management, and AI-powered tutoring sessions with Google Drive integration by implementing **FastAPI** with **SQLAlchemy ORM**.
- Integrated a **Gemini-powered chatbot** to deliver structured, context-aware responses to user queries.
- Boosted content retention for visual learners by automating short-form video generation using **MoviePy**, **ElevenLabs**, and **OpenAI** to convert notes into narrated content.

StockPicker | *Python, FastAPI, NGINX, Kafka*

- Built a real-time stock analytics platform with **FastAPI** and Kafka-based ingestion for processing market data.
- Implemented an **NGINX gateway** for request routing and rate limiting, deterministic sharding for scalable in-memory storage, and idempotent Kafka event handling.
- Developed unit, integration, and performance test suites validating **Kafka ingestion**, deterministic sharding, and API behavior under **787 requests/sec** with **7.9 ms p95 latency**.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, Kotlin, Bash, Typescript

Frameworks & Tools: Django, FastAPI, Spring Boot, Datadog, Docker, Kubernetes, Jira, Git/Github, Terraform

Databases: MongoDB, Postgres, SQLite